Gaussian conditioning below the mean-field scaling: a sharp fluctuation threshold

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Abstract

For a Bayesian tanh network with a Gaussian readout, the hidden feature kernel concentrates exponentially in width. When the scaling exponent satisfies $\gamma < 1$, this remains true after conditioning on a fixed output margin at every Boolean input and on noiseless training signs. For $0 < \gamma < 1$, a Gaussian moderate-deviation principle determines the leading margin energy. We ask whether it also determines the full conditioned posterior. For $0 < \gamma < 1/2$, we prove relative approximation of every full-margin orthant probability by a Gaussian with the deterministic finite-gain kernel, and hence total-variation approximation of every Boolean posterior. The threshold $1/2$ is sharp for a universal statement: with only three input bits and one hidden layer, an explicit pair of compatible labelings has identical Gaussian energy, but fourth-order feature fluctuations change their posterior odds by $\exp[(8586a^4/625)N^{2\gamma-1} + o(1 + N^{2\gamma-1})]$. At the original margin $a = 1/2$, these corrections change a prediction from less than $2/3$ confidence under the Gaussian comparison to more than $2/3$ confidence under the actual network, already at $\gamma = 1/2$. Thus Gaussian leading rates hold throughout the subcritical regime, while exact conditioned Gaussian behavior does not.

1 Model, conditioning, and the question

Let $\mathcal{X} = \{-1, 1\}^d$. Append a constant coordinate, $h^0(x) = (1, x)$, and use $L - 1$ hidden layers of common width N:

$$h^s(x) = \tanh(W^s h^{s-1}(x)), \quad 1 \le s < L, \qquad z_N(x) = W^L h^{L-1}(x).$$

There are no later biases and the readout is linear. All weights are independent centered Gaussians, with variances

$$\mathrm{Var}(W^1_{ij}) = N^{1-\gamma}/d, \qquad \mathrm{Var}(W^s_{ij}) = N^{-\gamma}\ (1 < s < L), \qquad \mathrm{Var}(W^L_i) = N^{-\gamma-1}.$$

These are the conventions in the original notes[1], continued to $\gamma < 1$. Let $Z_N = (z_N(x))_{x\in\mathcal{X}}$ and fix a margin $a > 0$, with $a = 1/2$ the original choice.

The prior is first conditioned on $|Z_N(x)| > a$ at every input. A training sample S then conditions on exact signs. If $\mathcal{G}_S$ denotes the compatible Boolean functions and $O_g = \{z : g(x)z(x) > a \text{ for all } x\}$, the posterior is exactly

$$\pi_N(g \mid S) = \frac{\Pr\{Z_N \in O_g\}}{\sum_{h\in\mathcal{G}_S} \Pr\{Z_N \in O_h\}}, \qquad g \in \mathcal{G}_S.$$

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[1] https://kaarelh.github.io/ai/nnbayes-generalization.pdf.

We also write A_g for the event $Z_N \in O_g$. Thus the all-input margin normalization cancels, but its restriction at unseen inputs remains in every orthant. For a target f, success at an input x means

$$\pi_N\{g : g(x) = f(x) \mid S\} > 2/3.$$

This is a sign posterior, not Gaussian regression with numeric observations of the latent outputs.

The variances prescribe a width-dependent schedule with fixed exponent γ: the normalized hidden-weight amplitude is $b(N) = N^{(1-\gamma)/2}$, and the output precision scale is $t(N) = N^{\gamma}$. These are the quantities denoted by b_N and t_N below; each is a scalar at a chosen width.

Every result below fixes d, L, a, γ before taking $N \to \infty$. Training labels may be any fixed consistent collection. Constants are not uniform as dimension grows. In particular, a comparison proved here cannot silently be used in a joint dimension–width limit or converted into a sample-complexity bound.

The exact Gaussian readout leaves just one source of non-Gaussian output behavior: the randomness of its feature covariance. A rare margin event can tilt that covariance by an amount tending to zero while still changing posterior odds by an unbounded factor. This is why convergence of the covariance and a Gaussian rate function are insufficient by themselves to identify a conditioned posterior.

Proof roadmap. We first control covariance fluctuations around the deterministic finite-gain recursion, including their effect after rare conditioning. A logarithmic sensitivity bound yields relative orthant probabilities when $N^{\gamma} = o(\sqrt{N})$. We then expand a strict-corner probability one order further and evaluate its fourth-moment correction in an eight-input example. The exact energy tie isolates that correction from the deterministic finite-gain bias.

2 Conditioned kernels and quantitative Gaussian comparison

Fix the input dimension d, depth L, and margin $a > 0$. Put $M = 2^d$, $b_N = N^{(1-\gamma)/2}$ and $t_N = N^{\gamma}$, and let Q_N be the empirical covariance matrix of the last hidden layer. All norms in this section are the maximum entry norm, unless another norm is indicated. Constants may depend on d, L, a; none of the statements is uniform in these parameters. The exact independent Gaussian readout gives

$$Z_N \mid Q_N \sim \mathcal{N}(0, Q_N/t_N).$$

For a full Boolean labeling g, set

$$O_g = \{z : g(x)z(x) > a \text{ for every } x\}.$$

For a collection S of observed labels, let $\mathcal{G}_S$ be the nonempty set of full labelings consistent with S, and let $A_S = \bigcup_{g \in \mathcal{G}_S} O_g$. Thus conditioning on A_S includes the margin requirement on every unobserved input.

2.1 The finite-gain center and conditional concentration

For a covariance matrix R, write

$$\Phi_b(R)_{xy} = \mathbb{E}[\tanh(bU_x)\tanh(bU_y)], \qquad U \sim \mathcal{N}(0, R).$$

Define deterministic matrices

$$R_0(x,y) = \frac{1 + x \cdot y}{d}, \qquad K_N^1 = \Phi_{b_N}(R_0), \qquad K_N^s = \Phi_{b_N}(K_N^{s-1}), \qquad \widehat{K}_N = K_N^{L-1}.$$

The distinction between $\widehat{K}_N$ and $\mathbb{E}Q_N$ matters: at depth greater than two the nonlinear covariance recursion does not commute with expectation. The limiting saturated kernel is

$$K^{(0)}(x,y) = \frac{1 + x \cdot y}{d+1}, \qquad K^{(s+1)} = \frac{2}{\pi} \arcsin K^{(s)}, \qquad K = K^{(L-1)},$$

where the arcsine is applied entrywise. This matrix is strictly positive definite. Indeed, the first-layer sign features associated to the nonparallel augmented inputs $(1, x)$ are linearly independent: crossing one of their distinct hyperplanes changes only its corresponding sign. Their Gram matrix is therefore positive definite. Subsequent arcsine maps preserve this property, by their positive power series and the Schur product theorem.

Lemma 2.1 (Concentration about the finite-gain kernel)**.** *For each fixed $\gamma < 1$, there are positive constants C, c, u_0 and N_0 such that, for $N \geq N_0$ and $0 < u \leq u_0$,*

$$\Pr\{\|Q_N - \widehat{K}_N\| > u\} \leq Ce^{-cNu^2}, \tag{1}$$

$$\|\widehat{K}_N - K\| \leq Cb_N^{-1}. \tag{2}$$

Proof. The diagonal variances of all limiting hidden kernels are one, and their distinct-coordinate correlations have absolute value strictly less than one. In fixed neighborhoods of these matrices, Φ_b has a Lipschitz constant independent of b. To verify the uniformity, each off-diagonal entry uses only a two-dimensional Gaussian marginal. Differentiate its density with respect to its covariance parameters. On a compact set of positive variances and correlations bounded away from ± 1, the L^1 norms of these density derivatives are bounded. The integrand $\tanh(bu)\tanh(bv)$ has absolute value at most one, so the resulting derivative bound is independent of b. The same one-dimensional argument handles diagonal entries. This argument does not require the full input covariance R_0 to be invertible.

Given the preceding hidden layer, its N successor rows are independent. Each entry of their empirical covariance averages N variables in $[-1, 1]$. Conditional Hoeffding bounds, a union bound over the M^2 entries, and the uniform Lipschitz estimate give (1) by induction over the fixed number of layers. At each stage one applies the previous-layer estimate with a constant multiple of u; exits from the fixed neighborhoods have probability at most Ce^{-cN} and are absorbed into the same bound.

For a centered Gaussian U of variance bounded below,

$$\mathbb{E}|\tanh(bU) - \operatorname{sign} U| \leq C/b.$$

The Gaussian sign-correlation formula and the same Lipschitz induction therefore give (2). □

Proposition 2.2 (Conditioning preserves the frozen kernel)**.** *For every fixed $\gamma < 1$,*

$$Q_N \longrightarrow K \quad \textit{in probability conditional on } A_S.$$

For $0 < \gamma < 1$, there is C_0 such that $\Pr(A_S) \geq e^{-C_0 t_N}$ for all sufficiently large N. Consequently, for fixed $\epsilon > 0$,

$$\Pr\{\|Q_N - K\| > \epsilon \mid A_S\} \leq C_\epsilon e^{-c_\epsilon N + C_0 t_N}.$$

More quantitatively, for a fixed sufficiently large constant A,

$$\Pr\left\{\|Q_N - K\| > Cb_N^{-1} + A\sqrt{t_N/N} \,\middle|\, A_S\right\} \leq C \exp[-(cA^2 - C_0)t_N].$$

These bounds may be chosen simultaneously for all observed-label collections S at the fixed input dimension.

Proof. Choose any $g \in \mathcal{G}_S$ and any fixed bounded box strictly inside O_g. On a sufficiently small fixed neighborhood of K, all covariance eigenvalues are bounded above and below. Integrating the density of $\mathcal{N}(0, Q_N/t_N)$ over the box gives a lower bound $ct_N^{M/2}e^{-Ct_N}$. That kernel neighborhood has probability tending to one, proving the asserted denominator bound. Dividing the bounds of Lemma 2.1 by this denominator proves the claims for positive γ. Here $t_N = o(N)$, and $b_N^{-1} = \sqrt{t_N/N}$ tends to zero.

If $\gamma = 0$, the normalized output converges to the nondegenerate Gaussian $\mathcal{N}(0, K)$ and the conditioning event has a positive limiting probability. If $\gamma < 0$, its normalized margin is $a\sqrt{t_N} \to 0$; the limiting conditioning event requires only the observed signs, again with positive Gaussian probability. The unconditional kernel concentration proves the corresponding conditional claim in both cases. □

2.2 What holds at every positive subcritical exponent

Set

$$I_K(z) = \frac{1}{2} z^T K^{-1} z, \qquad E_K(g) = \min_{z \in \overline{O}_g} I_K(z), \qquad E_S = \min_{g \in \mathcal{G}_S} E_K(g).$$

Each orthant has a unique minimizing vector z_g. Write

$$\mathcal{H}_S = \{g \in \mathcal{G}_S : E_K(g) = E_S\}, \qquad V_S = \{z_g : g \in \mathcal{H}_S\}.$$

Proposition 2.3 (Conditional latent concentration and unique minima)**.** *For every $0 < \gamma < 1$, the posterior concentrates on V_S. More precisely, for every $\delta > 0$ there is $\eta_\delta > 0$ such that*

$$\limsup_{N \to \infty} t_N^{-1} \log \Pr\{\operatorname{dist}(Z_N, V_S) \geq \delta \mid A_S\} \leq -\eta_\delta.$$

If $\mathcal{H}_S = \{g_*\}$, the posterior on full Boolean functions converges in total variation to δ_{g_*}. The fixed Gaussian $\mathcal{N}(0, K/t_N)$, conditioned on the same A_S, has the same limit. Thus their Boolean posteriors agree asymptotically throughout $0 < \gamma < 1$ whenever the compatible energy minimum is unique.

Proof. Couple the output to $Y_N = K^{1/2}\xi/\sqrt{t_N}$ by writing $Z_N = Q_N^{1/2}\xi/\sqrt{t_N}$, with ξ a standard independent M-dimensional Gaussian. Near the positive definite K, the matrix square root is Lipschitz. For fixed $\epsilon, \delta > 0$, kernel concentration and the Gaussian tail therefore bound

$$\Pr\{\|Z_N - Y_N\|_2 > \delta\} \leq C_\epsilon e^{-c_\epsilon N} + \Pr\{\|\xi\|_2 > \delta\sqrt{t_N}/(C\epsilon)\}.$$

After division by t_N in the logarithm, let $N \to \infty$ and then $\epsilon \downarrow 0$. Since $t_N = o(N)$, this proves exponential equivalence at speed t_N. Hence Z_N has the Gaussian large-deviation rate I_K.

The minimum rate on the strict orthant equals that on its closure: multiply its minimizing vector by $1 + \varepsilon$ and let $\varepsilon \downarrow 0$. The full-margin denominator consequently has logarithmic rate E_S. Coercivity of I_K and the finiteness of $\mathcal{G}_S$ give a strictly positive rate gap on the closed feasible set at distance at least δ from V_S. The large-deviation upper bound and the matching denominator lower bound give the displayed conditional estimate. If the minimizing labeling is unique, the same argument gives an exponential gap to every other compatible labeling. Apply the identical argument to Y_N. □

If several labelings minimize the energy, this proposition fixes their possible support but leaves their relative posterior weights undetermined. Those weights can decide whether a correct label has probability greater than 2/3. The next result controls them on a smaller exponent range.

2.3 Relative probabilities and total variation

Lemma 2.4 (Covariance sensitivity of an orthant probability)**.** *Let $p_g(Q,t) = \Pr\{\mathcal{N}(0, Q/t) \in O_g\}$. In a sufficiently small fixed convex neighborhood of K, uniformly over g and $t \geq 1$,*

$$|\log p_g(Q,t) - \log p_g(Q',t)| \leq Ct\|Q - Q'\|.$$

Proof. Uniformly over this covariance neighborhood, integrating over a fixed box inside each orthant gives $p_g(Q,t) \geq e^{-C_1 t}$, with constants chosen for all finitely many g. An unconditional Gaussian tail bound gives

$$\Pr\{\|Z\|_2^2 > u\} \leq Ce^{-ctu}$$

for u above a fixed constant. Divide by the preceding lower bound and integrate the resulting conditional tail, starting at a sufficiently large fixed u. It follows that $\mathbb{E}[\|Z\|_2^2 \mid Z \in O_g] \leq C$ uniformly for $t \geq 1$. Differentiating the Gaussian orthant integral along the segment from Q to Q' now gives a determinant derivative bounded by $C\|Q - Q'\|$ and a conditional quadratic-moment derivative bounded by $Ct\|Q - Q'\|$. Integration along the segment proves the claim. □

Theorem 2.5 (Quantitative Gaussian comparison)**.** *For $0 < \gamma < 1$, put $t = t_N$. There are positive constants C, c such that*

$$\max_g \left| \frac{\Pr\{Z_N \in O_g\}}{p_g(\widehat{K}_N, t)} - 1 \right| \leq C\frac{t}{\sqrt{N}} \exp(Ct^2/N) + C\exp(-cN + Ct)$$

for all sufficiently large N. In particular, if $0 < \gamma < 1/2$, the Boolean posterior of Z_N conditional on A_S is $o(1)$ in total variation from that of $\mathcal{N}(0, \widehat{K}_N/t_N)$ conditional on A_S. If $0 < \gamma < 1/3$, the same statement holds with K in place of $\widehat{K}_N$. These assertions include tied minimum-energy labelings.

Proof. Put $D = \|Q_N - \widehat{K}_N\|$ and restrict first to $D \leq u_0$, with u_0 chosen small enough for Lemma 2.4. On that event,

$$\left| \frac{p_g(Q_N, t)}{p_g(\widehat{K}_N, t)} - 1 \right| \leq e^{CtD} - 1.$$

Integration of the sub-Gaussian tail (1) yields

$$\mathbb{E}[(e^{CtD} - 1)\mathbf{1}_{D \leq u_0}] \leq C\frac{t}{\sqrt{N}} \exp(Ct^2/N).$$

For example, integrate $Ct \int_0^{u_0} e^{Ctu} \Pr(D > u)\, du$ and complete the square; enlarging the integral to $[0, \infty)$ gives the displayed bound. On $D > u_0$, the numerator is at most one, its probability is Ce^{-cN}, and the reference orthant probability is at least e^{-Ct}. This proves the relative-probability estimate, including the negligible subtraction needed in its lower bound.

When $t = o(\sqrt{N})$, the estimate tends to zero uniformly over all full sign patterns. Summing over $\mathcal{G}_S$ and dividing by the corresponding sums shows that conditional Boolean probabilities agree in total variation, uniformly over S. Finally, Lemmas 2.1 and 2.4 show that replacing $\widehat{K}_N$ by K changes each orthant probability by a factor $\exp[O(t/b_N)]$. This factor tends to one when $t/b_N = N^{(3\gamma-1)/2} \to 0$, proving the last assertion. □

The two comparisons address distinct errors. Empirical fluctuations of the hidden kernel lead to the sufficient condition $\gamma < 1/2$. Replacing the deterministic finite-gain kernel by its saturated limit introduces a separate bias, for which $\gamma < 1/3$ is a sufficient condition. Conditional kernel convergence and the leading Gaussian large-deviation rate remain valid over the entire range $\gamma < 1$.

3 A Gaussian rate function need not determine the conditioned posterior

Moderate deviations do justify a Gaussian rate function throughout $0 < \gamma < 1$. They do not, however, justify replacing the actual posterior by a conditioned Gaussian in total variation throughout this interval. The distinction already occurs with three input bits and one hidden layer. It affects the criterion that the correct label must have posterior probability greater than 2/3.

We first give the estimate that makes the counterexample rigorous. All vectors in the following lemma have a fixed dimension m.

Lemma 3.1 (Annealed Gaussian corner probabilities)**.** *For every N, let $H_{N,1}, \dots, H_{N,N}$ be independent copies of a random vector $H_N \in [-1,1]^m$. Suppose*

$$K_N = \mathbb{E} H_N H_N^T \longrightarrow K > 0, \qquad H_N \implies H.$$

Set $Q_N = N^{-1} \sum_i H_{N,i} H_{N,i}^T$, and, conditionally on the features, let $Z_N \sim \mathcal{N}(0, t_N^{-1} Q_N)$, where $t_N \to \infty$ and $t_N/N \to 0$. Fix $a > 0$ and $g \in \{-1,1\}^m$. Assume

$$g_i (K^{-1} g)_i > 0 \quad \text{for every } i.$$

Writing $v = K^{-1} g$, define

$$E_{N,g} = \frac{a^2}{2} g^T K_N^{-1} g, \qquad C_g = (2\pi)^{-m/2} (\det K)^{-1/2} \prod_{i=1}^m [a g_i v_i]^{-1}.$$

Then

$$\begin{aligned} \log \Pr\{ g_i Z_{N,i} > a \ \forall i \} = & -t_N E_{N,g} - \frac{m}{2} \log t_N + \log C_g \\ & + \frac{a^4 t_N^2}{8N} \operatorname{Var}((v^T H)^2) + o(1 + t_N^2/N). \end{aligned} \tag{3}$$

In particular, the leading correction to the deterministic Gaussian corner probability can grow without bound even though $t_N = o(N)$.

Proof. Write $D_g = \operatorname{diag}(g)$ and $E_g(Q) = a^2 g^T Q^{-1} g/2$. In a fixed neighborhood of K, the matrices Q are positive definite and all entries of $\beta_g(Q) = a D_g Q^{-1} g$ are bounded below by a positive constant. The change of variables $z = ag + D_g u / t_N$ in the Gaussian density gives

$$\begin{aligned} \Pr_Q\{ g_i Z_i > a \ \forall i \} = & (2\pi)^{-m/2} (\det Q)^{-1/2} t_N^{-m/2} e^{-t_N E_g(Q)} \\ & \times \int_{[0,\infty)^m} \exp\left\{ -\beta_g(Q)^T u - \frac{u^T D_g Q^{-1} D_g u}{2 t_N} \right\} du. \end{aligned}$$

The integral equals $\prod_i \beta_g(Q)_i^{-1} [1 + O(t_N^{-1})]$ uniformly in this neighborhood: use $0 \le 1 - e^{-r} \le r$ and the finite second moments of the product exponential density. Denote the resulting smooth positive prefactor by $C_g(Q)$.

Put $v_N = K_N^{-1} g$ and $\Delta_N = Q_N - K_N$. Matrix inversion has the uniform expansion

$$-t_N E_g(Q_N) = -t_N E_{N,g} + t_N \ell_N(\Delta_N) + O(t_N \|\Delta_N\|^2), \qquad \ell_N(\Delta) = \frac{a^2}{2} v_N^T \Delta v_N.$$

The linear statistic is the average of the bounded centered variables

$$Y_{N,i} = \frac{a^2}{2}\left[(v_N^T H_{N,i})^2 - \mathbb{E}(v_N^T H_N)^2\right].$$

Consequently, with $\theta_N = t_N/N$,

$$\begin{aligned}\log \mathbb{E} e^{t_N \ell_N(\Delta_N)} &= N \log \mathbb{E} e^{\theta_N Y_{N,1}} \\ &= \frac{t_N^2}{2N}\operatorname{Var}(Y_{N,1}) + O(t_N^3/N^2).\end{aligned}$$

All bounds here are uniform, since the $Y_{N,i}$ are uniformly bounded.

For completeness, the quadratic remainder is negligible at the claimed precision, even when t_N^2/N diverges. Tilt the independent feature rows by the density proportional to $\exp(\theta_N Y_{N,i})$. Under this product law their mean covariance differs from K_N by $O(\theta_N)$, and Hoeffding's inequality still gives, for a fixed number of matrix entries,

$$\Pr_{\theta_N}\{\|\Delta_N - \mathbb{E}_{\theta_N}\Delta_N\| > r\} \le Ce^{-cNr^2}.$$

Integrating this bound, and using $t_N/N \to 0$, yields

$$\log \mathbb{E}_{\theta_N} \exp\{Ct_N\|\Delta_N\|^2 + C\|\Delta_N\|\} = O\left(\frac{t_N^3}{N^2} + \frac{t_N}{N} + N^{-1/2}\right).$$

A matching lower estimate for the negative of the expression follows from Jensen's inequality. These bounds control both the inversion remainder and the smooth prefactor $C_g(Q_N)/C_g(K_N)$.

The complement of the fixed covariance neighborhood has probability $O(e^{-cN})$, by entrywise Hoeffding. Its contribution to the original orthant probability is at most this quantity. Inside the neighborhood the calculation gives a probability $\exp[-O(t_N) - O(\log t_N)]$; hence the omitted contribution is negligible. The same truncation under the tilted law is harmless, since the tilt is $o(N)$ and the features are bounded. We have thus obtained the expansion with $C_g(K_N)$, v_N, and H_N, and an error

$$O\left(t_N^{-1} + N^{-1/2} + t_N/N + t_N^3/N^2\right) = o(1 + t_N^2/N).$$

Finally $C_g(K_N) \to C_g$, and boundedness and weak convergence give $\operatorname{Var}((v_N^T H_N)^2) \to \operatorname{Var}((v^T H)^2)$. This proves the displayed claim. □

Theorem 3.2 (An actual-network counterexample to Gaussian posterior equivalence)**.** *Use the original architecture with $d = 3$, one hidden layer ($L = 2$), activation* tanh*, and any fixed margin $a > 0$. Order the cube inputs lexicographically, starting with $x_0 = (-1,-1,-1)$. Define*

$$g = (-1,1,1,-1,-1,-1,-1,1), \qquad h = (1,1,1,-1,-1,-1,-1,1).$$

Observe the seven common labels at every input other than x_0. Let p_N be the actual posterior probability of g, after the all-input-margin restriction and these observations. Let $\widetilde{p}_N$ be the corresponding probability for the deterministic Gaussian output vector $\mathcal{N}(0, N^{-\gamma}K_N)$, where K_N is the exact mean feature covariance at width N. Then

$$\widetilde{p}_N \longrightarrow \frac{189}{314} \qquad (0 < \gamma < 1),$$

whereas

$$p_N \longrightarrow \begin{cases} 189/314, & 0 < \gamma < 1/2, \\ \left[1 + \dfrac{125}{189} e^{-8586a^4/625}\right]^{-1}, & \gamma = 1/2, \\ 1, & 1/2 < \gamma < 1. \end{cases}$$

The same Gaussian limit holds if K_N is replaced by the limiting arcsine kernel K. Thus, for $1/2 \le \gamma < 1$, the actual Boolean posterior need not converge in total variation to either of these conditioned Gaussian posteriors.

For the original margin $a = 1/2$ and target g, the Gaussian learner's correct-label probability at the unseen input tends to $189/314 \simeq 0.602$, whereas the actual learner's probability tends to approximately 0.781 at $\gamma = 1/2$ and to one for $\gamma > 1/2$. The proposed $2/3$ confidence criterion therefore separates their predictions.

Proof. Let A be the 8×4 matrix with rows $(1, x)$, in the specified order; then $A^T A = 8I_4$. A feature row has distribution

$$H_N = \tanh\left(\frac{N^{(1-\gamma)/2}}{\sqrt{3}} AW\right), \qquad W \sim \mathcal{N}(0, I_4).$$

For $\gamma < 1$ it converges almost surely to $H = \operatorname{sign}(AW)$. Gaussian symmetry and oddness show, exactly at every finite width,

$$K_N = (q_N - 2c_N)I_8 + \frac{c_N}{2} AA^T,$$

where q_N is the diagonal feature covariance and c_N is the covariance between cube neighbors. The other off-diagonal entries are zero at Hamming distance two and $-c_N$ at distance three. Moreover $q_N \to 1$ and $c_N \to 1/3$, so

$$K = \frac{1}{3} I_8 + \frac{1}{6} AA^T, \qquad K^{-1} = 3I_8 - \frac{3}{10} AA^T.$$

The two functions satisfy

$$A^T g = (-2, -2, 2, 2)^T, \qquad A^T h = (0, -4, 0, 0)^T.$$

Their squared lengths are both eight and the displayed projection vectors both have squared length sixteen. Therefore $g^T K_N^{-1} g = h^T K_N^{-1} h$ exactly for all sufficiently large N; the finite-gain correction cannot break the energy tie.

Direct multiplication gives

$$v := K^{-1} g = \frac{3}{5}(-3, 5, 5, -7, -1, -3, -3, 5),$$
$$w := K^{-1} h = \frac{3}{5}(3, 3, 3, -7, -3, -3, -3, 7).$$

All signed gradient coordinates are strictly positive, and $g^T v = h^T w = 96/5$. Both orthant minimizers are therefore their corners ag and ah. The Gaussian corner prefactors have ratio

$$\frac{C_g}{C_h} = \frac{\prod_i h_i w_i}{\prod_i g_i v_i} = \frac{3^6\, 7^2}{3^3\, 5^3\, 7} = \frac{189}{125}.$$

The uniform Gaussian corner estimate in Lemma 3.1, before averaging over features, proves $\widetilde{p}_N \to 189/314$.

It remains to compute a fourth moment of the saturated feature row. Here this calculation is exact and finite. Signed permutations of the four coordinates of W give three orbits of possible threshold vectors H, of sizes $8, 64, 32$. To check completeness, first sort the absolute coordinates as $r_1 > r_2 > r_3 > r_4 > 0$. Only the signs of $r_1 - r_2 - r_3 - r_4$ and $r_1 - r_2 - r_3 + r_4$ are undetermined; their three feasible sign combinations give precisely these three orbits. The exceptional equalities have probability zero.

Let $P = AA^T/8$. The following table records finite averages within each orbit. Representatives are coefficient vectors W, not feature vectors. The column n_+ counts orbit elements with coordinates H_1, H_2, H_4, H_7 all positive, using zero-based cube indices.

representative W	size	$\|PH\|^2$	n_+	mean $(v^TH)^4$	mean $(w^TH)^4$
$(10, 3, 2, 1)$	8	8	1	1296/625	5184/625
$(5, 4, 2, 3/2)$	64	6	4	964224/625	1174176/625
$(4, 7/2, 3, 1)$	32	6	0	1174176/625	754272/625

Each entry can be checked by applying the $2^4 4!$ signed permutations to the representative, taking $\mathrm{sign}(AW)$, removing duplicates, and summing the indicated integer fourth powers. Gaussian invariance makes the distribution uniform within each orbit. To determine the three orbit probabilities p_1, p_2, p_3, note that

$$8p_1 + 6p_2 + 6p_3 = \mathbb{E}\|PH\|^2 = \mathrm{tr}(PK) = 20/3.$$

Thus $p_1 = 1/3$. The four rows of A with indices $1, 2, 4, 7$ are orthogonal, so their Gaussian projections have independent signs. Using the n_+ column,

$$\frac{1}{16} = \Pr\{H_1 = H_2 = H_4 = H_7 = 1\} = \frac{p_1}{8} + \frac{p_2}{16}.$$

It follows that $p_2 = p_3 = 1/3$. The table consequently gives

$$\mathbb{E}(v^TH)^4 = \frac{713232}{625}, \qquad \mathbb{E}(w^TH)^4 = \frac{644544}{625}.$$

The second moments are equal, both being $96/5$. Hence the difference of the two variances of squared projections is $68688/625$.

Apply Lemma 3.1 with $t_N = N^\gamma$. Since the two energies agree exactly, their prior log odds satisfy

$$\log \frac{\Pr(A_g)}{\Pr(A_h)} = \log \frac{189}{125} + \frac{8586a^4}{625} N^{2\gamma-1} + o(1 + N^{2\gamma-1}).$$

The seven observed labels leave exactly these two Boolean functions compatible, so these are also their posterior odds. The three stated limits follow directly. □

Remark 3.3 (What this counterexample does and does not show). The leading kernel margin energy remains correct at every $0 < \gamma < 1$: both functions in the example have the same energy, and the actual posterior selects one of these tied minimizers. The counterexample disproves full conditional-Gaussian equivalence, even with the exact finite-width mean covariance. It does not by itself establish a circuit prior or a general sample-complexity advantage. It shows that the leading kernel energy alone need not determine the posterior's confidence, and that fourth-order feature statistics can remain visible after rare-event conditioning.

4 Consequences for the circuit-prior research question

The strongest general statements proved here are as follows. For $\gamma \leq 0$, the limiting posterior is the Gaussian sign posterior, with a global margin at $\gamma = 0$ and no margin at $\gamma < 0$. For $0 < \gamma < 1/2$, the deterministic finite-gain Gaussian approximates every full Boolean posterior, including tied energy minima. The sufficient range for replacing that kernel by its saturated limit is $0 < \gamma < 1/3$; optimality of the latter threshold is not asserted. For every $0 < \gamma < 1$, the saturated Gaussian rate and concentration on compatible energy minimizers remain valid. A unique compatible minimum gives the same limiting Boolean posterior throughout this entire interval. With ties, Theorem 3.2 disproves a universal Gaussian-posterior replacement for every $1/2 \leq \gamma < 1$.

These comparisons concern probabilities themselves. A strict $2/3$ confidence indicator is discontinuous at probability exactly $2/3$; total-variation convergence alone does not settle that boundary case. The counterexample stays strictly away from the boundary on both sides and therefore gives an unambiguous separation under the stated strict confidence convention.

A companion paper, *Small circuits can have exponentially large prior complexity below the mean-field scaling*, proves a separate conclusion throughout $\gamma < 1$: full parity has a polynomial-size, polynomial-weight tanh circuit but exponential prior complexity on the natural scale of the regime. That prior separation does not imply exponential labeled-sample complexity. Conversely, the present non-Gaussian correction does not restore a polynomial circuit prior. It resolves a different issue: leading kernel energies need not determine how the Bayesian learner weights equally costly explanations.

A statistical question not addressed here is whether the actual all-input-margin-conditioned posterior needs exponentially many examples to predict a k-bit XOR correctly with probability strictly greater than $2/3$ on most inputs. For full parity ($d = k$), the companion paper *Exponential parity sample complexity below the mean-field scaling* proves an $\exp(\Omega(k))$ lower bound in the width-first limit. For sparse parity, $\exp(\Omega(k))$ bounds are proved for $\gamma < 0$ and for $0 < \gamma < 1$; the growing-sparse case at $\gamma = 0$ and the natural $\exp\{\Omega(k \log(ed/k))\}$ scale for $0 < \gamma < 1$ at fixed depth remain open. Neither the prior-mass separation nor the usual posterior-mean regression argument proves such a lower bound. The counterexample also means that a proof covering the whole subcritical interval cannot simply replace every posterior by a Gaussian posterior and ignore its tied minima.

Verification. The accompanying script, `verify-counterexample.py`, checks the finite orbit table, projection identities, strict corner gradients, energy equality, Gaussian prefactor ratio, and fourth-moment coefficient using exact rational arithmetic. The asymptotic estimates and the completeness of the orbit classification are proved in the text.